

Report

My research focused on modelling the behaviour of speakers of minority languages in a society dominated by one other language. Close to home this is the current situation of an Gaeilge in Ireland, but there are many other comparable cases across the world often because of colonialism. My research sought to model the societal and individual factors that influence what language that a minority language speaker will use in the Irish context (whether someone who can speak Irish will speak it, and why).

Past Research

An extensive amount of past research was reviewed for this paper, but in the interests of brevity 3 studies of most importance can be identified.

“An application of an analogue of the partition function to the evolution of diglossia” by Wyburn and Hayward. This study focuses on diglossia (a state of relatively stable bilingualism in which speakers use different languages for different functions within society) and uses a cellular automaton to model a community of speakers in a bilingual society.

“Agent-based models of language competition” by Castello et al.

This study uses an agent-based model to analyse the effects of language prestige and language loyalty (how unwilling a speaker is to change to another language). It explores how social networks also influence the spread of a dominant majority language at the cost of a minority one.

“Proficiency, attitude and conventions in minority languages” by Laruelle et al.

This study focuses on a conversation game where two modelled agents interact with one another. Each agent has its own proficiency in a language and attitude towards using that language. There is also a 3rd factor: language conventions. These represent the societal expectation of what language is to be used in a particular interaction. E.g a majority biased convention will always favour the majority language unless both speakers have a relative preference for the minority one.

The Model

A decision was made to build a multi agent model using the framework of proficiency, attitude and conventions outlined in Laruelle et al but extended to a social network of many agents rather than simple 1 on 1 interactions.

Data for proficiency was sourced from Irish census data, ordered by electoral division. There are over 3400 electoral divisions in Ireland, giving highly granular data. Census data contains questions about how often speakers use Irish, which was used to inform proficiency information. Attitude information was not as straightforward to find, being sourced from published surveys.

By the end of the research scholarship period, a model had been creating using python. Several different graphs were used to generate networks, including a connected caveman graph, LFR benchmark graph, WS small world graph and various other random graphs. The model could be called for any electoral division.

The project was incomplete by the end of the scholarship period, and I intend to work on it more and eventually seek to have it published. Policy changes could be modelled using the model where, for example, a campaign to increase pride in Irish may increase attitudes, education may increase

proficiency, and encouraging Irish speakers to connect socially may be modelled by adding connections between the nodes representing the speakers.

Below I have included some test outputs for a randomly chosen electoral district in Ireland. (please excuse the difficult-to-read graphs. On the left is a graph showing the proficiency values (blue = high proficiency, red=low). On the right is a graph showing the attitude values. The red connections between nodes show cases where a conversation between two speakers is in Irish. The top set of nodes is a minority-biased convention, the middle is a majority biased, and the final set is a neutral convention

